Software Engineering Assignment

Project: Hospital Management System (HMS)

1. Introduction

A Hospital Management System (HMS) is a comprehensive software suite designed to manage all aspects of a hospital's operations. This includes patient registration, electronic health records (EHR), billing, pharmacy inventory, laboratory results, and staff scheduling. Because this system handles life-critical data and sensitive personal information, any failure can have severe consequences, making thorough planning and risk assessment essential.

2. Selected SDLC Model

Spiral Model

3. Justification

The Spiral model is particularly suited for high-stakes, complex systems like an HMS for several reasons:

Intense Risk Analysis: In a hospital setting, risks—such as data breaches or system downtime—must be mitigated. The Spiral model includes a dedicated risk assessment phase in every "loop," ensuring potential issues are identified and solved early.

Complexity Management: An HMS is massive, involving various departments (Radiology, ICU, Billing). The Spiral model allows developers to build the system in segments, handling the immense complexity one step at a time.

User Involvement and Feedback: Since doctors and administrative staff will use the system, the model's iterative nature allows for constant prototyping. Feedback from medical professionals can be incorporated at the end of each cycle.

Flexibility and Customization: As healthcare regulations (like HIPAA or GDPR) change, the Spiral model provides the flexibility to adjust requirements during the development process without disrupting the entire project structure.

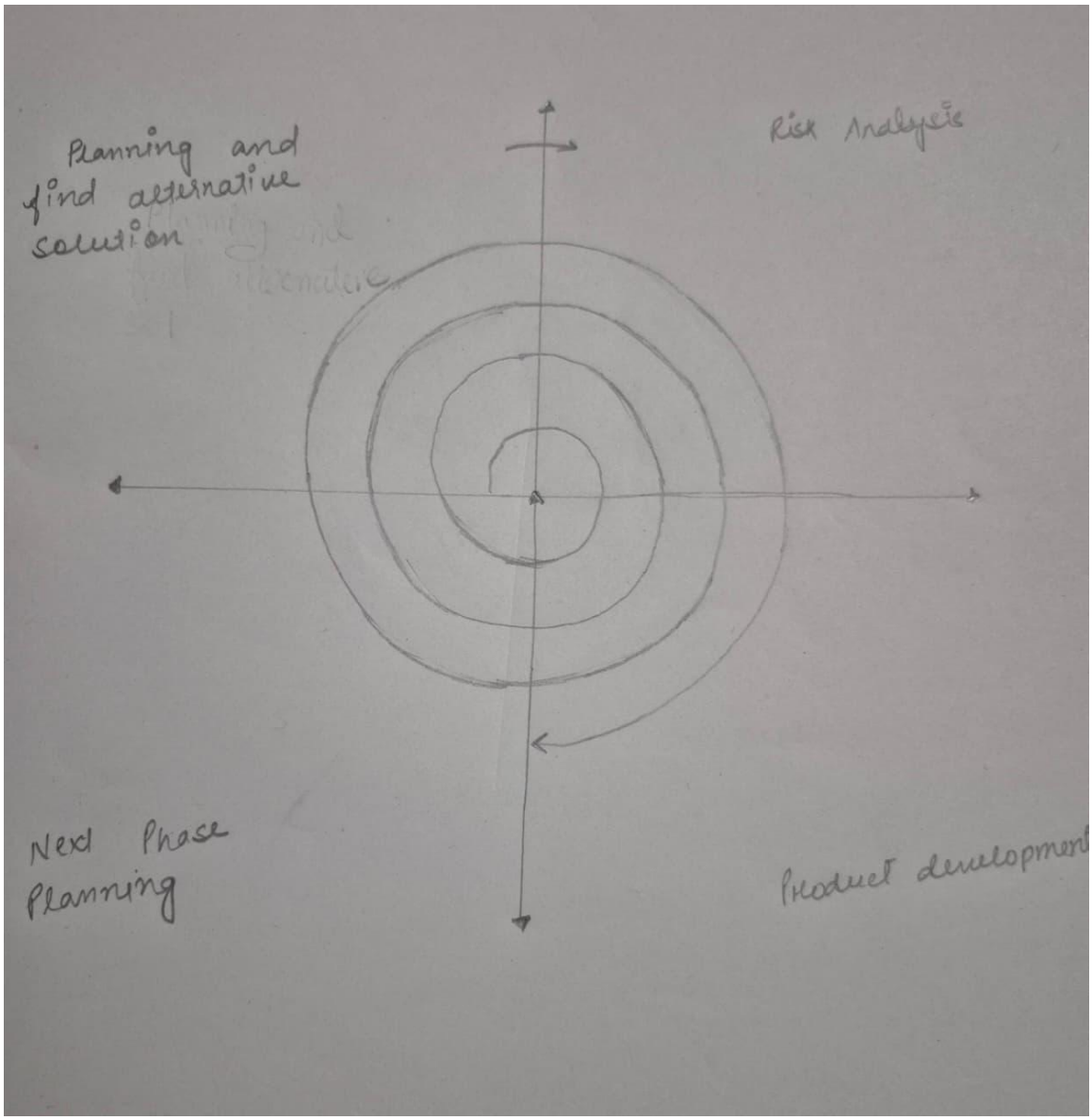
High Reliability: Because the system undergoes rigorous testing and evaluation at the end of every quadrant, the final product is much more stable and reliable than those built with faster, less structured methods.

4. Comparison with Other Models

Waterfall Model: This is too risky for an HMS. If a critical flaw in the data architecture is discovered during the testing phase (near the end), fixing it would be astronomically expensive and time-consuming. It lacks the iterative feedback needed for such a complex

user interface
Agile Model: While Agile is great for speed, it can sometimes lack the formal documentation and rigorous upfront risk analysis required for healthcare compliance. In a hospital, you cannot "move fast and break things"—everything must be meticulously documented and verified

5. Spiral Model Diagram



6. Conclusion

The **Spiral Model** is the most logical choice for a Hospital Management System because it prioritizes **risk management** above all else. In an environment where software errors can

impact patient health and data privacy, the Spiral model's disciplined, iterative approach ensures that the system is both robust and compliant with strict medical standards.